W03 Revised Critical Requirements

ENGSE601 • THE ONE Bakery • Proposed revision; owner approval pending

ร่าง requirement และ AC จาก S1-B + code review; ไม่พบ formal SRS IDs/version และไม่มี stakeholder approval. DRAFT IDs ด้านล่างเป็นรหัสชั่วคราวเพื่อ trace ในเอกสารนี้เท่านั้น.

Review scope and source

Feature from S1-B: การบันทึกคำสั่งซื้อ; additional implementation concerns from routes/orders.js, middleware/auth.js, database/schema.sql and README-FIX12.md.

DRAFT-ORD-01 — Order recording and total

Field	Record
Source statement	S1-B p.4: เมื่อสั่งซื้อ ระบบต้องบันทึกคำสั่งซื้อและคำนวณยอดรวมให้ถูกต้อง. Formal SRS ID/version not supplied.
Issue	ยังไม่ระบุ quantity integer/boundaries, error behavior และ partial-write expectations; route ไม่เห็น explicit integer validation.
Proposed wording	เมื่อรับคำสั่งซื้อ ระบบต้องตรวจว่ามีรายการสินค้าอย่างน้อยหนึ่งรายการ สินค้าทุกชิ้น active และ quantity เป็นจำนวนเต็มบวก จากนั้นคำนวณ total จากราคาที่เก็บฝั่ง server และบันทึก order/items/status อย่างสอดคล้องกัน.
Status	Proposed / Not Ready until owner confirms rules.
Evidence plan	API response + orders/order_items in test DB; currently Not Executed.

Acceptance criteria (proposed)

ID	Input/context	Expected observable result	Type
AC-ORD-01	valid active products, positive integer quantities, multiple items	server total equals sum of authoritative price × quantity; items store correct unit_price; new order begins pending_review	Normal
AC-ORD-02	empty items or missing/inactive product	client error; no partial order/item records	Negative
AC-ORD-03	quantity 0, negative, or 1.5	client error before DB write, not generic server error	Boundary

DRAFT-ORD-02 — Status transition and authorization

Field	Record
Source	routes/orders.js: FLOW, permitted roles, POST /:id/status, POST /:id/decision; middleware/auth.js. Formal requirement absent.
Issue	Status route has role/sequence guards; POST /:id/cancel does not visibly use auth middleware or check order ownership in handler. Need design decision for customer authorization.
Proposed wording	Only authorized roles may change staff workflow statuses and only to the next permitted state. All status changes must write the history record. Cancellation must verify caller authority for that order and must not rely on knowledge of an order ID.
Status	Proposed / Not Ready until customer identity/authorization approach is confirmed.
Evidence plan	API tests for no credential, wrong role, skipped state and authorized cancellation; currently Not Executed.

Acceptance criteria (proposed)

ID	Input/context	Expected observable result	Type
AC-ORD-04	Kitchen attempts pending_review → picked_up	request denied; order status and history remain unchanged	Negative/auth
AC-ORD-05	caller lacks permission for an order cancellation	request denied and original status preserved	Security
AC-ORD-06	authorized actor sends next valid transition	only next status recorded and from/to audit history exists	Normal/audit

DRAFT-INV-01 — Recipe-based ingredient deduction

Field	Record
Source	routes/orders.js: consumeInventoryForOrder; database/schema.sql: recipes, inventory, inventory_movements; README-FIX12.md. Formal requirement absent.
Issue	README-FIX12 notes orders without recipes can transition but create no movement. Recipe coverage and unit consistency are not verified.
Proposed wording	When production moves from in_production to ready_for_packing, compute ingredients from recipe × order quantity, lock/check stock, deduct and record movements atomically. If stock is insufficient, roll back status and all stock changes. Never deduct the same order twice.
Status	Proposed / Conditional; actual DB/recipe coverage not verified.
Evidence plan	Test-DB snapshots and movement rows for sufficient/insufficient/repeat transitions; Not Executed.

Acceptance criteria (proposed)

ID	Input/context	Expected observable result	Type
AC-INV-01	recipe exists and stock is sufficient	stock decreases by recipe quantity × order quantity and movement records link order/ingredient/unit	Normal
AC-INV-02	required ingredient exceeds stock	transaction rolls back; status and all stocks remain unchanged; no false movement	Negative/boundary
AC-INV-03	same order consumption attempted twice	no second deduction/movement; invalid state transition is rejected	Idempotency

Open questions and handoff

Question/blocker	Decision owner	Impact	Status
Which Proposal/SRS and version is source of truth?	Project owner/team	Needed for traceability and formal baseline	Not Ready
What rounding and quantity formats are allowed?	Project owner/shop representative	Financial and item correctness	Open
What authorization method proves right to cancel an order?	Project owner/developer	Security control design	Open / high
Which active products have recipes, and what are the units?	Inventory owner	Stock deduction correctness	Not Executed
Who reviews and approves revised ACs?	Requirement owner	AI/team drafts are not stakeholder decisions	Not Available

AI declaration: AI helped structure proposed wording and acceptance criteria from source code and the supplied worksheet. Team must verify, revise or reject each proposal in AI-LOG.md. No test results or screenshots were generated by AI.

Do not relabel these DRAFT IDs as official SRS IDs until mapped to an actual requirements source.